

Лабораторная работа №3

Вариант №3

Листинг библиотеки классов (dll):

```
1 using System;
2
3 namespace dll
4 {
5     public class Machine
6     {
7         public string Metal { get; set; } = "None";
8         public int Age { get; set; } = 0;
9         public bool Break { get; set; } = false;
10
11         public void Repair()
12         {
13             if (Break)
14             {
15                 Break = false;
16                 Console.WriteLine("Car is fixed");
17             }
18
19             else
20                 Console.WriteLine("Car is in good condition");
21
22         }
23
24         public void setAge(int new_age)
25         {
26             Age = new_age;
27             Console.WriteLine($"Set age: {Age}");
28         }
29
30         public void ShowInfo()
31         {
32             string Status = Checking();
33             Console.WriteLine($"\\nMachine age: {Age}\\nStatus: {Status}\\nMetal
: {Metal}");
34         }
35
36         public string Checking()
37         {
38             if (!Break)
39                 return "serviceable";
40             else
41                 return "break";
42         }
43     }
44
45     public class Car : Machine
46     {
47         public int id { get; set; } = 0;
48         public string Model { get; set; } = "None";
49         public int Places = 0;
50         public Engine engine = new Engine();
```

```

51     public bool Drive = false;
52
53
54     public void DriveToggle()
55     {
56         if (Drive) { Drive = false; }
57         else { Drive = true; }
58
59         if (Drive && !Break)
60         {
61
62             if (!engine.Work)
63             {
64                 engine.Start();
65             }
66
67             Console.WriteLine("Drive");
68         }
69
70         else if (Break)
71         {
72             Drive = false;
73             Console.WriteLine("Car is broken!");
74         }
75         else
76         {
77             if (engine.Work)
78             {
79                 engine.Stop();
80             }
81             Console.WriteLine("Stop drive");
82         }
83     }
84
85
86     public void Honk() { Console.WriteLine("HONK!"); }
87
88     public void ReplaceEngine()
89     {
90         if (engine.Work) { engine.Stop(); }
91
92         Console.WriteLine("Model engine: ");
93         string model = Console.ReadLine();
94
95         Console.WriteLine("Cylinders engine: ");
96         int cylinders = int.Parse(Console.ReadLine());
97
98         Console.WriteLine("HorsePower engine: ");
99         int horse_power = int.Parse(Console.ReadLine());
100
101         Engine new_engine = new Engine { model_engine = model, Cylinders
= cylinders, HorsePower = horse_power };
102         engine = new_engine;
103     }
104
105

```

```

106     public void modelEngine()
107     {
108         Console.WriteLine(engine.model_engine);
109
110     }
111
112     public Car CreateCar()
113     {
114         Car newCar = new Car();
115
116         Console.WriteLine("Model: ");
117         newCar.Model = Console.ReadLine();
118
119         Console.WriteLine("Places: ");
120         newCar.Places = int.Parse(Console.ReadLine());
121
122         Console.WriteLine("--Machine Properties--");
123         Console.WriteLine("Metal: ");
124         newCar.Metal = Console.ReadLine();
125
126         Console.WriteLine("Age: ");
127         newCar.Age = int.Parse(Console.ReadLine());
128
129         Console.WriteLine("Break (true/false): ");
130         newCar.Break = bool.Parse(Console.ReadLine());
131
132         Console.WriteLine("--Engine Properties--");
133         /*newCar.engine = new Engine();*/
134         newCar.engine.createEngine();
135
136         return newCar;
137
138     }
139
140 }
141
142 public class Engine
143 {
144     public int id { get; set; } = 0;
145     public string model_engine { get; set; } = "None";
146     public int Cylinders { get; set; } = 0;
147     public int HorsePower { get; set; } = 0;
148     public bool Work = false;
149
150     public Engine createEngine()
151     {
152
153         Console.WriteLine("Model: ");
154         string Model_engine = Console.ReadLine();
155         Console.WriteLine("Cylinders: ");
156         int cylinders = int.Parse(Console.ReadLine());
157         Console.WriteLine("HorsePower: ");
158         int horsePower = int.Parse(Console.ReadLine());
159
160         return new Engine { Cylinders = Cylinders, HorsePower =
horsePower, model_engine = Model_engine };

```

```

161     }
162
163     public void Start()
164     {
165         if (!Work) { Work = true; } else { Console.WriteLine("The engine
is already running"); }
166     }
167
168     public void Stop()
169     {
170         if (Work) { Work = false; } else { Console.WriteLine("The engine
is no longer running"); }
171     }
172
173
174     public void Upgrade(int extraHorsPower)
175     {
176         HorsePower = extraHorsPower;
177     }
178
179 }
180 }

```

Листинг веб-сервиса (asmx.cs):

```

1 using dll;
2 using System;
3 using System.ComponentModel;
4 using System.Configuration;
5 using System.Data;
6 using System.Data.SqlClient;
7 using System.Web.Services;
8
9 namespace freakWeb
10 {
11
12     [WebService(Namespace = "http://tempuri.org/")]
13     [WebServiceBinding(ConformsTo = WsiProfiles.BasicProfile1_1)]
14     [System.ComponentModel.ToolboxItem(false)]
15
16
17     public class WebService1 : System.Web.Services.WebService
18     {
19
20         private string connectionString = ConfigurationManager.ConnectionStrings["db"
].ConnectionString;
21
22         [WebMethod]
23         public int CreateCars(string metal, int age, string model, bool @break, int
places)
24         {
25             using (SqlConnection connection = new SqlConnection(connectionString))
26             {
27                 connection.Open();
28                 string sqlInsertMachine = "INSERT INTO Machines (Metal, Age, [Break])
VALUES (@Metal, @Age, @Break); SELECT SCOPE_IDENTITY();";

```

```

29         int machineId;
30         using (SqlCommand command = new SqlCommand(sqlInsertMachine,
connection))
31         {
32             command.Parameters.Add("@Metal", SqlDbType.NVarChar, 50).Value =
metal;
33             command.Parameters.Add("@Age", SqlDbType.Int).Value = age;
34             command.Parameters.Add("@Break", SqlDbType.Bit).Value = @break;
35             machineId = Convert.ToInt32(command.ExecuteScalar());
36         }
37
38
39         string sql = "INSERT INTO Cars (Model, Places, Id) VALUES (@Model,
@Places, @Id); SELECT SCOPE_IDENTITY()";
40
41         using (SqlCommand command = new SqlCommand(sql, connection))
42         {
43             command.Parameters.Add("@Id", SqlDbType.Int).Value = machineId;
44             command.Parameters.Add("@Model", SqlDbType.NVarChar, 50).Value =
model;
45             command.Parameters.Add("@Places", SqlDbType.Int).Value = places;
46             command.ExecuteNonQuery();
47             return machineId;
48         }
49     }
50 }
51
52 [WebMethod]
53 public BindingList<Car> GetAllCars()
54 {
55     BindingList<Car> cars = new BindingList<Car>();
56
57     using (SqlConnection connection = new SqlConnection(connectionString))
58     {
59         connection.Open();
60         string sql = "SELECT * FROM Cars";
61
62         using (SqlCommand command = new SqlCommand(sql, connection))
63         {
64             using (SqlDataReader reader = command.ExecuteReader())
65             {
66                 while (reader.Read())
67                 {
68                     Car car = new Car
69                     {
70                         id = (int)reader["Id"],
71                         Model = reader["Model"] == DBNull.Value ? null : (
string)reader["Model"],
72                         Places = reader["Places"] == DBNull.Value ? 0 : (int)
reader["Places"],
73                         Drive = reader["Drive"] == DBNull.Value ? false : (
bool)reader["Drive"],
74
75                     };
76                     cars.Add(car);
77                 }

```

```

78         }
79     }
80 }
81 return cars;
82 }
83 }
84 }

```

Листинг приложения Windows Forms

```

1 using Forms.ServiceReference1;
2 using System;
3 using System.ComponentModel;
4 using System.Drawing;
5 using System.IO;
6 using System.Windows.Forms;
7 using System.Xml.Serialization;
8
9 namespace Forms
10 {
11     public partial class Form1 : Form
12     {
13         private BindingList<Car> car_list = new BindingList<Car>();
14         public int count_car = 1;
15         public Form1()
16         {
17             InitializeComponent();
18             // dataGridView1.DataSource = car_list;
19             comboBox1.Items.AddRange(new object[] { "id", "Metal", "Age", "
Break" });
20         }
21
22         private void Search_Click_1(object sender, EventArgs e)
23         {
24             string searchValue = textBox1.Text;
25             string chouse = comboBox1.Text;
26
27             foreach (DataGridViewRow row in dataGridView1.Rows)
28             {
29                 row.DefaultCellStyle.BackColor = Color.White; // сброс цвета
30
31                 if (row.Cells[chouse].Value?.ToString() == searchValue)
32                 {
33                     row.DefaultCellStyle.BackColor = Color.Tan;
34                 }
35             }
36         }
37
38         private void ClearSearch_Click(object sender, EventArgs e)
39         {
40             foreach (DataGridViewRow row in dataGridView1.Rows)
41             {
42                 row.DefaultCellStyle.BackColor = Color.White;
43             }
44         }
45     }

```

```

46 private void CreateCustomCar_Click_1(object sender, EventArgs e)
47 {
48     bool Break = inputBreak.Checked;
49
50     string Model = inputModel.Text;
51     string Metal = inputMetal.Text;
52     int Age = int.Parse(inputAge.Text);
53     int places = int.Parse(inputPlaces.Text);
54
55     using (WebService1SoapClient client = new WebService1SoapClient()
56     {
57         client.CreateCars(Metal, Age, Model, Break, places);
58     })
59     {
60     private void SaveCarToXml(BindingList<Car> car, string filePath)
61     {
62         XmlSerializer serializer = new XmlSerializer(typeof(BindingList<
63 Car>));
64         using (StreamWriter writer = new StreamWriter(filePath))
65         {
66             serializer.Serialize(writer, car);
67         }
68     }
69     private BindingList<Car> LoadCarFromXml(string filePath)
70     {
71         XmlSerializer serializer = new XmlSerializer(typeof(BindingList<
72 Car>));
73         using (StreamReader reader = new StreamReader(filePath))
74         {
75             return (BindingList<Car>)serializer.Deserialize(reader);
76         }
77     }
78     private void Grid2xml_Click_1(object sender, EventArgs e)
79     {
80         SaveFileDialog saveDialog = new SaveFileDialog();
81         if (saveDialog.ShowDialog() == DialogResult.OK)
82         {
83             SaveCarToXml(car_list, saveDialog.FileName);
84         }
85     }
86
87     private void Xml2grid_Click_1(object sender, EventArgs e)
88     {
89         OpenFileDialog openFileDialog = new OpenFileDialog();
90         if (openDialog.ShowDialog() == DialogResult.OK)
91         {
92             car_list = LoadCarFromXml(openDialog.FileName);
93             dataGridView1.DataSource = car_list;
94         }
95     }
96
97     private void CreateTestCar_Click_1(object sender, EventArgs e)
98     {

```

```

99         Car car = new Car
100     {
101         id = count_car,
102         Model = "Tesla",
103         Metal = "Aluminum",
104         Age = 2,
105         Break = false,
106     };
107     car_list.Add(car);
108     count_car++;
109 }
110
111
112 private async void Sql2grid_Click(object sender, EventArgs e)
113 {
114     WebService1SoapClient client = new WebService1SoapClient();
115
116     ServiceReference1.GetAllCarsResponse response = await client.
117     GetAllCarsAsync();
118
119     BindingList<ServiceReference1.Car> cars = new BindingList<
120     ServiceReference1.Car>(response.Body.GetAllCarsResult);
121
122     dataGridView1.DataSource = cars;
123
124     if (client.State != System.ServiceModel.CommunicationState.
125     Faulted)
126     {
127         client.Close();
128     }
129 }

```

Листинг SQL скриптов:

- cars.sql

```

1 CREATE TABLE [dbo].[Car] (
2     [Id] INT NOT NULL PRIMARY KEY,
3     [Model] NVARCHAR(10) NULL DEFAULT 'None',
4 );

```

- engine.sql

```

1 CREATE TABLE [dbo].[Engines] (
2     [Id] INT NOT NULL PRIMARY KEY IDENTITY(1,1),
3     [ModelEngine] NVARCHAR(10) NULL DEFAULT 'None',
4     [Cylinders] INT NULL DEFAULT 0,
5     [HorsePower] INT NULL DEFAULT 0,
6     [Work] BIT NULL DEFAULT 0
7 );

```


- machine.sql

```
1 CREATE TABLE [dbo].[Machines]
2 (
3     [Id] INT NOT NULL PRIMARY KEY IDENTITY(1,1),
4     [Metal] NVARCHAR(10) NULL DEFAULT 'None',
5     [Age] INT NULL DEFAULT 0,
6     [Break] BIT NULL DEFAULT 0
7 )
```

Результаты работы программы: